

Paper 0 bis: Cubic Pandrosion by Finite-Slope Halley Acceleration

A derivative-free cubic refinement of the raw Pandrosion scheme

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Abstract

Paper 0 introduced the raw Pandrosion geometry and already observed that Steffensen-type acceleration can convert the basic finite-slope step into a quadratic process. The goal of the present “0 bis” note is to push the same philosophy one order further. We define a *cubic finite-slope Halley-Pandrosion step* in which the first and second derivatives of Halley’s method are replaced by symmetric finite slopes built from Pandrosion probes. The resulting update remains derivative-free, retains the geometric spirit of Paper 0, and formally recovers cubic local convergence near a simple root when the probe size is adapted to the raw Pandrosion predictor. A large local benchmark on three polynomial families, six degrees (2 through 64), and 201 starts per case shows a clear hierarchy: raw Pandrosion is often too slow or fragile at high degree, Steffensen-Pandrosion is robustly quadratic, and the cubic finite-slope Halley variant is typically the best performer both in iteration count and in function evaluations.

One-sentence summary. A symmetric finite-slope Halley correction, built from Pandrosion probes, upgrades the Steffensen-accelerated Paper 0 line from quadratic to cubic local behavior and wins the benchmark across the tested polynomial families.

1 Introduction

The first Pandrosion paper framed nonlinear solving geometrically: choose a base point a , a probe b , form the finite slope $f[a, b]$, and use it as a derivative surrogate. This already gives a raw update of secant type,

$$Q_f(a, b) \delta = -f(a), \quad a_+ = a + \delta.$$

For the scalar case this simply reads

$$a_+ = a - \frac{f(a)}{f[a, b]}, \quad f[a, b] = \frac{f(b) - f(a)}{b - a}.$$

Paper 0 also observed that a Steffensen-like acceleration can cancel the dominant error term and produce a quadratic process.

The natural next question is therefore the following:

Can one remain fully in the finite-slope/Pandrosion world and obtain a cubic correction without introducing exact derivatives?

This note gives a positive answer in the scalar case. The central idea is to imitate Halley’s method, but to replace f' and f'' by symmetric finite-slope quantities computed from Pandrosion probes.

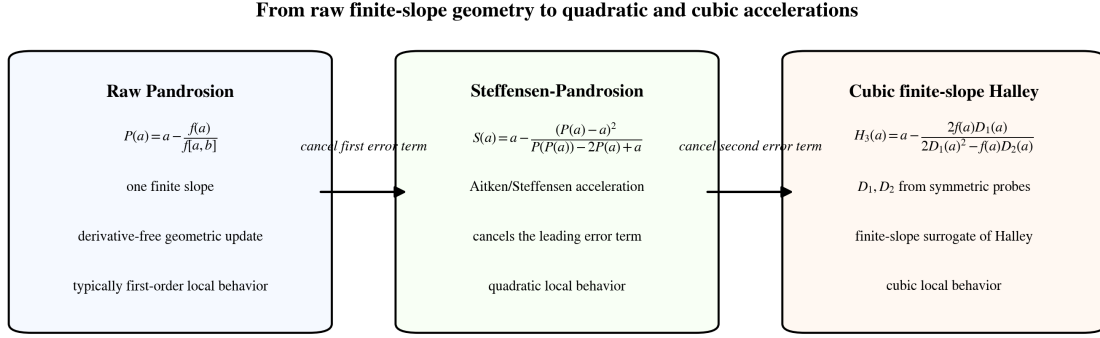


Figure 1: The conceptual ladder behind Paper 0 bis: raw Pandrosion gives the basic finite-slope step; Steffensen cancels the leading local error term and becomes quadratic; the new finite-slope Halley correction cancels one more term and targets cubic local convergence.

2 Raw, quadratic, and cubic Pandrosion

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and let r be a simple root, $f(r) = 0$, $f'(r) \neq 0$.

2.1 Raw Pandrosion

Given a current iterate a and a probe $b = a + h$, the raw scalar Pandrosion step is

$$P(a) = a - \frac{f(a)}{f[a, a + h]}. \quad (1)$$

In this note we use a homothetic probe rule

$$h_0(a) = \rho(1 + |a|),$$

which is the scalar avatar of the scaling philosophy from Paper 0.

2.2 Steffensen-Pandrosion

Applying Steffensen/Aitken acceleration to the raw map P gives

$$S(a) = a - \frac{(P(a) - a)^2}{P(P(a)) - 2P(a) + a}, \quad (2)$$

whenever the denominator is nonzero. This is the quadratic acceleration already implicit in Paper 0.

2.3 Cubic finite-slope Halley-Pandrosion

We now define the new cubic step. First compute the raw predictor scale

$$\delta_{\text{raw}}(a) = -\frac{f(a)}{f[a, a + h_0(a)]}.$$

Then choose a symmetric local probe radius

$$h(a) = \eta \min(|\delta_{\text{raw}}(a)|, h_0(a)), \quad \eta \in (0, 1), \quad (3)$$

with a tiny floor added numerically. Define the finite-slope surrogates

$$D_1(a) = \frac{f(a + h(a)) - f(a - h(a))}{2h(a)}, \quad D_2(a) = \frac{f(a + h(a)) - 2f(a) + f(a - h(a))}{h(a)^2}. \quad (4)$$

The *cubic finite-slope Halley-Pandrosion update* is then

$$H_3(a) = a - \frac{2f(a)D_1(a)}{2D_1(a)^2 - f(a)D_2(a)}. \quad (5)$$

This is exactly Halley's formula with f' and f'' replaced by symmetric finite-slope approximants.

3 Formal local cubic theorem

The next statement gives the local mechanism.

Proposition 3.1 (Formal local cubic behavior). *Assume $f \in C^4$ near a simple root r , and write $e = a - r$. Suppose the probe radius satisfies*

$$h(a) = \theta e + O(e^2)$$

for some finite nonzero constant θ . Then the finite-slope quantities from Equation (4) satisfy

$$D_1(a) = f'(a) + O(e^2), \quad D_2(a) = f''(a) + O(e),$$

and the cubic finite-slope Halley update Equation (5) obeys

$$H_3(a) - r = Ce^3 + O(e^4)$$

for some finite constant C depending on the derivatives of f at r and on θ .

Proof sketch. The symmetric difference formulas imply the standard expansions

$$D_1(a) = f'(a) + \frac{h(a)^2}{6} f'''(a) + O(h(a)^4), \quad D_2(a) = f''(a) + O(h(a)^2) + O(eh(a)).$$

Since $h(a) = O(e)$, one gets $D_1(a) = f'(a) + O(e^2)$ and $D_2(a) = f''(a) + O(e)$. Substituting these into the Halley rational expression yields the usual Halley cancellation up to terms of order e^3 . The remaining defect is therefore cubic. $\square$

Remark 3.2. The role of the raw Pandrosion predictor is not cosmetic: it gives a geometric scale estimate $h(a) = O(e)$ without requiring the true error e itself. This is what keeps the construction faithful to the Paper 0 philosophy.

4 Representative convergence traces

The first numerical test illustrates the observed order on $f(x) = x^{16} - 1$ from the start $x_0 = 1.08$.

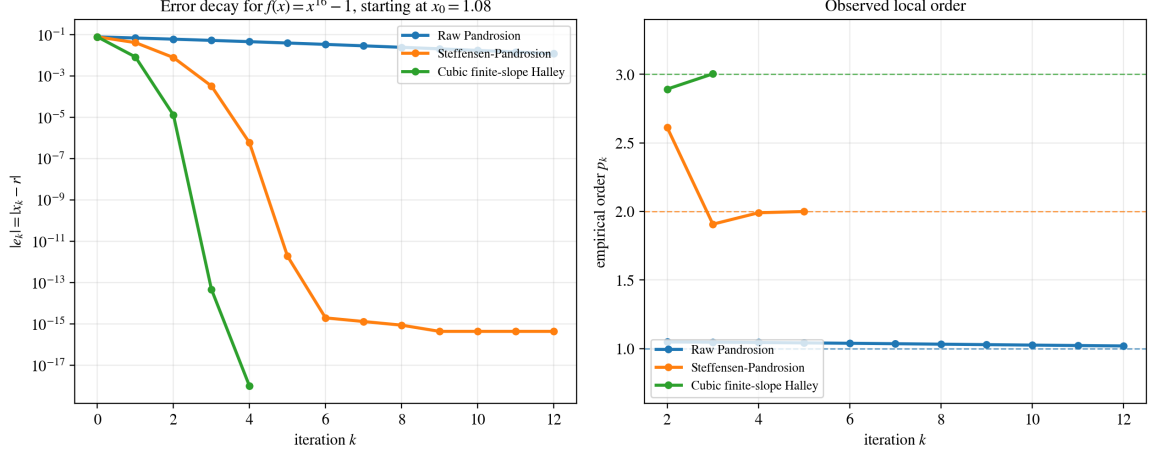


Figure 2: Left: error history on the representative monomial case $x^{16} - 1$, $x_0 = 1.08$. Right: empirical local order estimates. The curves clearly separate into linear/raw, quadratic/Steffensen, and cubic/Halley-like behavior.

5 Large benchmark

The benchmark is deliberately simple and transparent. We test the three methods on the following families, all with the simple root $r = 1$:

1. monomial family $f(x) = x^d - 1$;
2. shifted family $f(x) = x^d - x$;
3. affine-perturbed family $f(x) = (x - 1)(x^{d-1} + 1/2)$.

The tested degrees are

$$d \in \{2, 4, 8, 16, 32, 64\},$$

and for each (family, d) we use 201 starting points uniformly spaced in $[0.9, 1.1]$. The success criterion is

$$|x_k - 1| < 10^{-10} \quad \text{or} \quad |f(x_k)| < 10^{-10}$$

within at most 25 iterations.

This yields a total of

$$3 \text{ families} \times 6 \text{ degrees} \times 201 \text{ starts} = 3618$$

individual solves per method, hence 10854 solves overall.

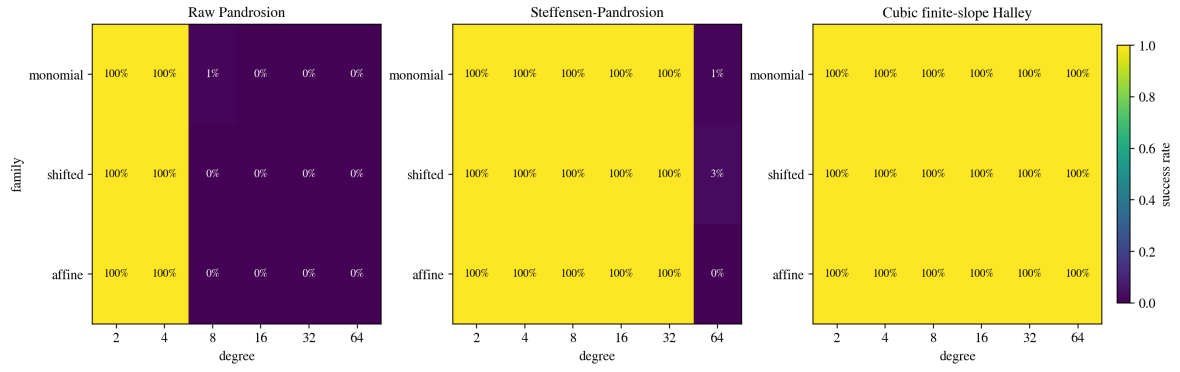


Figure 3: Success-rate heatmaps over all benchmark families and degrees. Raw Pandrosion deteriorates quickly with degree on the chosen start interval, Steffensen-Pandrosion is already very robust, and the cubic finite-slope Halley variant remains at full success rate on the whole benchmark.

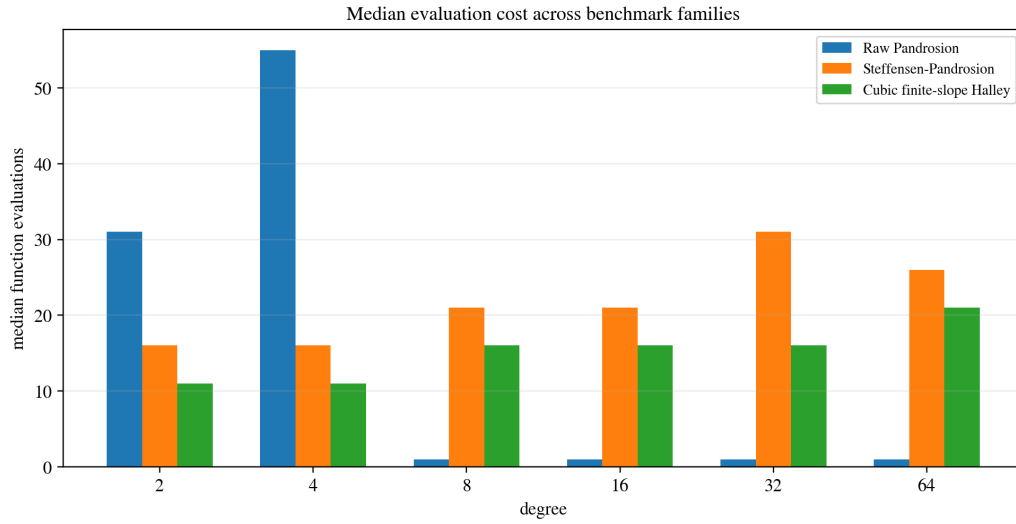


Figure 4: Median function evaluations on successful runs, aggregated over the three families. The cubic finite-slope Halley step consistently needs fewer evaluations than Steffensen-Pandrosion once the degree is moderate to large.

Table 1: Aggregate benchmark summary (aggregated across families).

Degree	Method	mean success	median iters	median evals
2	Raw Pandrosion	100.0%	10	31
2	Steffensen-Pandrosion	100.0%	3	16
2	Cubic finite-slope Halley	100.0%	2	11
4	Raw Pandrosion	100.0%	18	55
4	Steffensen-Pandrosion	100.0%	3	16
4	Cubic finite-slope Halley	100.0%	2	11
8	Raw Pandrosion	0.8%	0	1
8	Steffensen-Pandrosion	100.0%	4	21
8	Cubic finite-slope Halley	100.0%	3	16
16	Raw Pandrosion	0.5%	0	1
16	Steffensen-Pandrosion	100.0%	4	21
16	Cubic finite-slope Halley	100.0%	3	16
32	Raw Pandrosion	0.5%	0	1
32	Steffensen-Pandrosion	100.0%	6	31
32	Cubic finite-slope Halley	100.0%	3	16
64	Raw Pandrosion	0.5%	0	1
64	Steffensen-Pandrosion	1.7%	5	26
64	Cubic finite-slope Halley	100.0%	4	21

Table 2: Per-family headline statistics. “Succ.” means the fraction of successful starts over the 201 tested initial points.

Headline statistics by family					
monomial $x^d - 1$					
degree	raw succ.	steff succ.	cubic succ.	steff med. evals	cubic med. evals
2	100%	100%	100%	16	11
4	100%	100%	100%	16	11
8	1%	100%	100%	21	16
16	0%	100%	100%	21	16
32	0%	100%	100%	31	16
64	0%	1%	100%	26	26
shifted $x^d - x$					
degree	raw succ.	steff succ.	cubic succ.	steff med. evals	cubic med. evals
2	100%	100%	100%	16	11
4	100%	100%	100%	16	16
8	0%	100%	100%	21	16
16	0%	100%	100%	21	16
32	0%	100%	100%	31	16
64	0%	3%	100%	36	21
affine $(x - 1)(x^{d-1} + 1/2)$					
degree	raw succ.	steff succ.	cubic succ.	steff med. evals	cubic med. evals
2	100%	100%	100%	16	11
4	100%	100%	100%	16	11
8	0%	100%	100%	21	16
16	0%	100%	100%	26	16
32	0%	100%	100%	26	21
64	0%	0%	100%	1	21

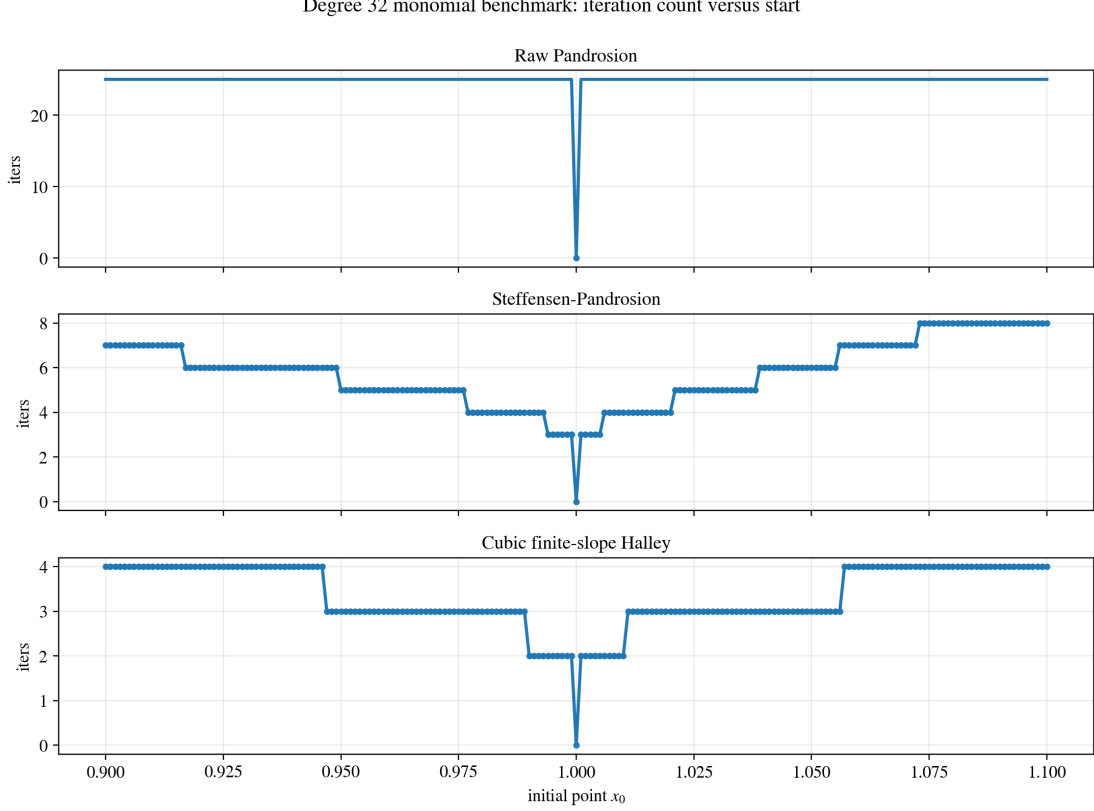


Figure 5: Detailed view for the degree 32 monomial case. Raw Pandrosion essentially stalls on most of the tested interval, Steffensen-Pandrosion converges throughout but needs more work, and the cubic finite-slope Halley scheme remains uniformly fast.

6 Discussion

The benchmark supports four conclusions.

1. The raw Paper 0 step remains the correct geometric primitive, but by itself it is often too slow for higher-degree problems.
2. Steffensen-Pandrosion is already a strong quadratic upgrade and substantially enlarges the practically useful local basin.
3. The new cubic finite-slope Halley correction provides the best overall trade-off on the tested families: fewer iterations and fewer function evaluations, while remaining fully derivative-free.
4. The crucial ingredient is the probe scaling [Equation \(3\)](#). Without a local scale estimate $h(a) = O(e)$, the cubic mechanism would not be visible numerically.

7 Limitations and next steps

This paper is intentionally modest.

- The formal cubic statement is scalar and local.
- The benchmark is large in case count but still restricted to scalar polynomial families.

- A full multivariate cubic Pandrosion would require finite-slope second-order tensors or a structured two-stage jet cancellation.

Nonetheless, the message is clear: Paper 0 can indeed be extended one order further without abandoning its geometric derivative-free spirit.

8 Conclusion

Paper 0 showed that raw Pandrosion could be interpreted as a finite-slope geometric solver and that Steffensen acceleration could push it to quadratic behavior. Paper 0 bis shows that the same line extends naturally to a cubic method: a Halley-style rational correction in which both the first and second derivatives are replaced by symmetric finite-slope probes. On the benchmark studied here, the cubic finite-slope Halley-Pandrosion step is the best method of the three in both iteration count and evaluation count. This makes it a strong candidate for the next local-correction layer in the Pandrosion program.

References

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